REDUCING GREENHOUSE GAS (GHG) EMISSIONS THROUGH SOLAR CELL PHOTOVOLTAIC PLANNING AT AL-JIHAD ISLAMIC BOARDING SCHOOL IN SURABAYA

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ABSTRACT

Currently, 80% of conventional energy is used to meet the needs of industry and the general public. Using new, renewable energy from the sun is an effort to mitigate climate change by reducing greenhouse gases (GHG). Al-Jihad Boarding School, an Islamic educational institution with around 1,200 students, is one of the biggest consumers of conventional energy. This study aims to plan photovoltaic solar cells, calculate the amount of power that can be generated, calculate the amount of GHG emission reduction from photovoltaic solar cells and the costs required for installation at Al-Jihad Islamic Boarding School. The planning results were analyzed by adjusting the selection of a 12 KVA inverter, 44 polycrystalline photovoltaic solar cells, and other complementary materials such as cables, MCBs and supports. The amount of power generated from photovoltaic solar cells at the Al-Jihad Islamic Boarding School in Surabaya is 12 kWh. Climate change mitigation efforts by reducing GHG emissions through solar cell photovoltaic planning at the Al-Jihad Islamic Boarding School in Surabaya can reduce CO_2 greenhouse gases by 1,200.5 kg, NH_4 30.013 kg, and N_2O 0.019 kg. The cost required for the installation of photovoltaic solar cells is Rp. 209,850,400.

Keywords: *Greenhouse Gas; Al-Jihad Islamic Boarding School; Solar Cell Photovoltaic*

1. INTRODUCTION

Energy is the most important element for humans to survive and is involved in every aspect of life. Energy has become an important concern in all countries in recent decades. Human lifestyle in modern times has a very close relationship with the quantity of energy and its quality (Nurjaman & Purnama, 2022).

Energy consumption in Indonesia still depends on non-renewable energy as an energy source. Indonesia has many conventional energy sources, such as oil, coal, and natural gas. However, this source will also run out, coupled with the fact that using this energy source also contributes to greenhouse gas emissions and causes climate change (Hasan *et al.*, 2012). This high energy consumption will continue to increase yearly, with the share of energy from oil and natural gas and in 2025 amounting to 49% and 2050, 39% of the total energy needs in Indonesia (Setyono & Kiono, 2021).

The depletion of oil and gas reserves and the rapid growth in conventional energy, as well as continuous energy consumption, force us to find renewable energy sources, such as solar, wind, biomass and hydropower, to support economic development in the future (Handayani & Ariyanti, 2012). One of the abundant sources of new renewable energy in Indonesia, as a tropical country, is energy from the sun. The advantages are that it is non-polluting, has no moving parts that can break easily, requires little maintenance and minimal supervision, and has a lifespan of 20-30 years with relatively low operational costs (Bagher, *et al.*, 2015).

Using new, renewable energy from the sun is one of the efforts to mitigate climate change that can be carried out in many sectors. Climate change mitigation is carried out as an effort to prevent climate change. One mitigation strategy is to reduce Green House Gases (GHG). One effort can be made by installing solar panels/solar cell photovoltaics. The use of solar cells (solar panels) can function to minimize the effects of Carbon dioxide (CO_2) emissions (Miloudi, 2013; Maybe *et al.*, 2020).

Solar energy, as sustainable energy whose availability will never run out, can be used as a basic material for converting sunlight into electricity using solar cells/PV. Solar cells or panels are a device framework that can convert sunlight into electrical energy using a photovoltaic process (Purwanto *et al.*, 2020). In general, solar cells are divided into three types, namely monocrystalline, polycrystalline and thin film, where monocrystalline solar cells have higher efficiency than polycrystalline solar cells and thin film cells (Budiyanto & Fadliondi, 2017). Solar panels as alternative energy are also more efficient in investment and operational costs, when compared to generators as a power source (Purwoto *et al.*, 2018).

A study conducted at the IPB Darmaga campus showed that using solar cells/solar panels is estimated to reduce carbon dioxide (CO_2) emissions by 651 tons within 25 years (Boer & Setiadi, 2016). Calculation of CO_2 emissions as one of the Green House Gas (GHG) emissions by using solar panels only produces emissions of 20.31%; in other words, it can reduce CO_2 gas emissions in the air by up to 79.69% (Sulistiawati & Yuwono, 2019).

Solar cells can be applied anywhere, as long as sufficient sunlight daily. This is the basis that Indonesia has great potential for utilizing solar cells because it is in a tropical region that is exposed to sunlight for 8-12 hours. The design and application of solar cells can also be carried out in urban areas, namely in residential areas or residential complexes; in industrial areas such as factories; and other places such as office buildings, hotels, malls, schools, universities, hospitals, airports, stations, libraries, museums, amusement parks (recreation), and so on (Ramadhan & Rangkuti, 2016).

Al-Jihad Islamic Boarding School Surabaya is one of the large Islamic boarding schools in Surabaya. The students and female students generally come from UINSA students due to their position close to this agency. This Islamic boarding school has an area consisting of 3 Islamic boarding school buildings, one mosque and one orphanage. The use of solar cells at the Al-Jihad Islamic boarding school, apart from being an effort to mitigate climate change by reducing GHG emissions, also aims to introduce the concept of renewable energy and as a long-term investment in economic aspects in its planning. This cottage has a roof that is large enough to be used as a planning area for photovoltaic solar cells to produce electrical energy from sunlight. Therefore, to researchers

2. METHODS

2.1. Types of Research

The type of research that will be applied is quantitative descriptive. Quantitative selection comes from data collection to analysis of data displayed in numbers. Meanwhile, descriptive is in the form of a description of the condition and energy needs of the Islamic boarding school and an explanation of the resulting data.

2.2. Research Sites

The research will be conducted at the Al-Jihad Islamic boarding school in Surabaya.

2.3. Data Collection

This research will have variables that include:

a) Technical aspects: Includes supporting facilities for solar cell photovoltaic planning at the Al-Jihad Islamic Boarding School in Surabaya.
b) Environmental aspects: Covers the conditions for planning the construction and operation of photovoltaic solar cells at the Al-Jihad Islamic Boarding School in Surabaya regarding environmental pollution.
c) Economic aspect: Includes the economic potential of funding solar cell photovoltaic planning and calculating future profits.
d) Institutional aspect: This includes the institutional structure at the Al-Jihad Surabaya Islamic Boarding School which will later be responsible for planning the development and operation of photovoltaic solar cells.

2.4. Data Collection Technique

The data collection techniques that will be implemented in this research are as follows:

a) Literature Study

   Literature study in involves and collecting data in the form of written documents,

 P-ISSN:2356-3109 E-ISSN: 2356-3117

reports of similar research results, and supporting data during research such as books, journals and articles.

b) Observation and Interview

Observation is making observations of the object to be studied. Interviews will be conducted by asking more in-depth questions regarding conditions and important information so that people understand the object of the research.

c) Data Calculations

Calculate the energy requirements needed at the Al-Jihad Islamic Boarding School in Surabaya. With these calculations, further analysis is carried out to determine the capacity of the photovoltaic solar cell.

2.5. Data Analysis Technique

Based on the description of the data collection techniques above, the following data analysis techniques will be obtained:

a) Calculation of Energy Requirements

Calculating daily energy requirements is needed to determine the capacity and units of solar cell photovoltaic planning. Initial calculations are carried out on each electronic item which will later be supplied with electrical energy and can be formulated according to Darma (2017) as follows:

$$Energy\ per\ unit\ (WH) = Volt\ unit\ (Watt)\ x\ Usage\ (hours) \quad (1)$$

After that, the total energy needs of the Islamic boarding school for one full day can be calculated by adding up the energy per unit of electronic goods, such as lights, fans, refrigerators, and so on.

$$Total\ daily\ energy\ requirements\ (KWH) = epu\ 1\ +\ eput\ 2\ +\ ... \quad (2)$$

b) Determine the solar cell photovoltaic unit

Some important information must be known before planning a photovoltaic solar cell: energy requirements per day (KWH), maximum power contract with PLN (Watts), and the electrical network can be 1 phase or 3 phase.

After that, a solar cell photovoltaic unit can be determined with several parts:

1) Inverter selection

Explanation: The inverter is required to be 1 phase. The inverter must provide power output by the maximum power contract with PLN (Watts). The inverter must be given energy input from solar panels as much as the maximum power contract with PLN (Watts).

2) Selecting the size and number of solar panels

The number of solar panels can be assumed, provided the maximum power contract with PLN (watts) is the same.

Determining the solar cell photovoltaic unit can adjust the goods to the marketing area in general, according to the calculations and energy requirements required.

c) Calculation of GHG Emission Reduction by using solar cells

The reduction in GHG emissions can be determined based on the fuel consumed per type. This refers to the method issued in 2006 by the Intergovernmental Panel on Climate Change (IPCC) regarding Guidelines for National Greenhouse Gas Inventories.

Based on the Directorate General of Electricity (2018), power plants are categorized as stationary. Therefore, the general equation applied in calculating GHG emissions is as follows:

$$GHG\ Emissions\ (kg) = E.Consumption\ (kg)\ x\ Factor\ (kg/TJ) \quad (3)$$

According to the IPPC default, the emission unit is applied per unit of energy consumed (kg.GRK/TJ). Because of this, energy consumption must first be converted into TJ (Terra Joule) energy units. The conversion equation is explained as follows:

$$Energy\ Consumption\ (TJ) = Consumption\ x\ Cal\ Value\ (TJ) \quad (4)$$

d) Calculation of Funding Requirements

The costs that must be paid to build a photovoltaic solar cell will adjust to the price of goods on the market. If it is known then the total cost of all components and the funding requirements for solar cell photovoltaic planning at the Al-Jihad

Islamic boarding school in Surabaya are known. In addition, there is a BEP (Break Even Point) calculation, which is a position that explains expenditure and income at the same point. With the following formula:

$$BEP = \frac{Total\ cost\ of\ expenditure}{Watt\ PLN\ x\ Effective\ Hours\ x\ Unit\ Price} \quad (5)$$

3. RESULTS AND DISCUSSION

3.1. Al-Jihad Islamic Boarding School Information

Al-Jihad Surabaya Islamic Boarding School which is located on Jl. Jemursari Utara III/9 Surabaya, is one of the large Islamic boarding schools in Surabaya. The santri and female students generally come from student circles. The Al-Jihad Surabaya Islamic boarding school has 7 buildings with an average of 1-7 floors. Based on the results of interviews, it was found that the number of male students was 300, but during the pandemic only 120 remained. Meanwhile, the number of female students is 800, with a total during the pandemic of only 440. With this data, it is calculated that the population ratio of the Al-Jihad Surabaya Islamic boarding school is 40-55%.

Al-Jihad Islamic boarding school has a total of 12,000 watts of electrical power from each building. In the last three months, the total electricity bill for the Al-Jihad Islamic Boarding School in Surabaya is more than IDR 62,624,216. This shows that the use of electrical energy in the Islamic boarding school area is very large.

3.2. Average Electrical Energy Requirements

The average daily electricity requirement calculation is 110,594.4 watts/day at the Al-Jihad Islamic Boarding School in Surabaya. This is calculated based on the number of electronic devices, length of use, and wattage of the item. Lamps, rice cookers, Sanyo pumps, and chargers frequently consume much electrical power.

3.3. Design of Solar Cell Photovoltaic Units

The design of photovoltaic solar cells at the Al-Jihad Islamic Boarding School in Surabaya uses a 1-phase electricity system. As a result, in the future Islamic boarding schools will try to produce their energy fully, without having to ask for electricity assistance from PLN. A single-phase electrical network system has an output of 2 electrical conductor channels, namely neutral (N) and phase (L). It has a large output voltage of around 220 volts – 240 volts, this is by Indonesian standards.

Selecting an inverter requires 1 phase. The inverter must provide power output using the maximum power contract with PLN (Watts). The inverter must be given energy input from solar panels as much as the maximum power contract with PLN (Watts), namely 12,000 Watts. The GoodWe Smart DT series is specifically designed for single-phase roof systems; the 12kW, 15kW, 17KW, and 20kW power ranges are ideal for small commercial use. Two integrated MPPTs allow two-string input from different roof orientations. SDT series inverters are small, light, and easy to install. This inverter is suitable for outdoor and indoor installations and offers quiet operation due to fanless natural convection cooling.

The choice of size and number of solar panels can be calculated, provided that the maximum power contract with PLN (watts) and the previous determination of the inverter is 12,000 Watts. Obtained a solar cell with a power output of 275 Watts. Based on calculations, it was found that the number of solar cells was 44. Then, it is connected in series so that the Total Voltage and Total Current calculations are 1667.6 Volts and 9.35 A respectively.

Table 1. Solar Cell Specifications

No	Element	Information
1.	Electrical Parameter at STC	
	a. Modul type	YLxxxP-29b 1500V
	b. Power output	275 Watt
	c. Power output tolerences	O/+S
	d. Module efficiency	16,8 %
	e. Voltage at Pmax	31,0 Volt
	f. Current at Pmax	8,90 Ampere
	g. Open-circuit voltage	37,9 Volt
	h. Short-circuit current	9,35 Ampere
2.	Operating Conditions	
	a. Max. system voltage	1500 V_{DC}
	b. Max. series fuse rating	15 Ampere

 P-ISSN:2356-3109 E-ISSN: 2356-3117

No	Element	Information
	c. Liming reverse current	15 Ampere
	d. Operating temperature range	-40°C to 85°C
	e. Max. static load, front	5400 Pa
	f. Max. static load, back	2400 Pa
	g. Max. hailstone impact (d/v)	25 mm/23 mm/s
3.	Construction Materials	
	a. Front cover (material/ thickness)	Low-iron tempered glass/ 3,2 mm
	b. Cell (quantity/ material/ dimension/ number of busbars)	60/ multicrystalline silicon/ 156,75 mm x 156,75 mm (+/-0,25)/ 4 or 5
	c. Frame (material)	Anodized alumunium alloy
	d. Junstion box (protection degree)	> IP67
	e. Cable (length/ cross-sectional area)	1000 mm/ 4 mm^2
	f. Plug connector (type/ protection degree)	MC4/IP68 or Amphenol H4/ IP68 of Fostfor SIKE6/ or Renhe RH05-8/ IP67
4.	General Characteristics	
	a. Dimensions (L/W/H)	1650 mm/992 mm/35 mm
	b. Weight	18,5 kg
5.	Packaging Specifications	
	a. Number of modules per pallet	30
	b. Number of pallets per 40’ container	28
	c. Packaging box dimensions (L/W/H)	1700 mm/1135 mm/1165 mm
	d. Box weight	588 kg

(Source: Yingli, 2021)

The scheme for converting solar panels into electricity is carried out by an inverter that converts the DC voltage from the solar panels to AC voltage. This also synchronizes the GRID voltage or what can be called PLN voltage, and then distributes it to the load. The inverter towards the load will continue the solar energy absorbed by the solar panels, if the load requires it. For example, if the energy received by the solar panels turns out to be greater than the load requirements, then the excess energy has the potential to be sent to the PLN network. If something like that happens, the Islamic Boarding School will experience a surplus of electrical energy. This surplus energy is recorded in the form of KWH meters as energy IMPORT. Meanwhile, for example, if the energy received by the solar panels turns out to be less than what is needed by the load, then the PLN supplies the lack of energy when the Islamic boarding school experiences a shortage of electrical energy. This minus energy is recorded on the KWH meter as EXPORT energy. The inverter will work automatically to regulate the load's energy needs.

The plan for installing solar cells at the Al-Jihad Islamic Boarding School in Surabaya is designed to utilize roof and rooftop space due to limited new land. This is also an effort to design efficiently and not take up much space.

Figure 1. Al-Jihad Islamic Boarding School Rooftop Area

Several rooftops and places that can be used as planning locations for solar cell installation are the roof of the Usman Building (160 m^2), the rooftop of the Umar Building (76 m^2), the At-Tin Building (220 m^2), and the Bung Hatta Building (130 m^2). The location is taken based on sufficient and spacious roof space availability.

3.4. Calculation of GHG Emission Reduction by Using Solar Cells

The reduction in GHG emissions can be determined based on the amount of fuel consumed per type, which is calculated using the formula (3). This refers to the method issued in 2006 by the Intergovernmental Panel on Climate Change (IPCC) regarding Guidelines for National Greenhouse Gas Inventories. Power plants and the like fall into the non-moving (stationary) category. According to the IPPC default, the emission unit is applied per unit of energy consumed (kg.GHG/TJ). Because of this, energy consumption must first be converted into TJ (Terra Joule) energy units and calculated using formula (4). Based on these formulas, it can be calculated as follows:

Is known:

a) The Islamic boarding school's electrical energy source is PLTU Paiton, Probolinggo Regency, East Java.
b) Paiton PLTU uses coal as its source of electrical energy.
c) The calorific value of coal is 28.2 TJ/Gg
d) Coal emission factors for several gases include:

   CO_2 = 94,600 kg GHG/TJ

   CH_4 = 1 kg GHG/TJ

   N_2O = 1.5 kg GHG/TJ

   (Directorate General of Electricity, 2018)
e) The average energy consumption of Islamic boarding schools is 75,911.25 watts/day = 75.9 kWh
f) 1 kWh = 0.59 kg Coal (Riadessy, 2015)

Coal burning calculations are as follows:

Coal burning (kg) = 75.9 x x 0.59 kg

Coal burning (kg) = 44,781 kg

Coal burning (kg) = 4.5 x 10^{-5} Gg

The energy consumption calculation obtained is as follows:

Energy Consumption (TJ)

= 4.5 x 10^{-5} Gg x 28.2 TJ/Gg

= 0.01269 TJ ≈ 0.013 TJ

So the calculation of Green House Gas (GHG) emissions is as follows:

1) GHG CO_2

   GHG emissions CO_2 (kg)

   = 0.013 TJ x 94,600 kg GHG/TJ

   = 1,200.5 kg GHG CO_2
2) GHG CH_4

   GHG emissions CH_4 (kg)

   = 0.013 TJ x 1 kg GHG/TJ

   = 0.013 kg GHG CH_4
3) GHG N_2O

   GHG emissions N_2O (kg)

   = 0.013 TJ x 1.5 kg GHG/TJ

   = 0.019 kg GHG N_2O

So it can be assumed that if the Al-Jihad Surabaya Islamic Boarding School uses solar energy which is converted into electrical energy through solar cell photovoltaic technology, then the Islamic Boarding School can reduce greenhouse gases by 1,200.5 kg of CO_2, 3,013 kg of NH_4, and 0.019 kg of N_2O. kg.

3.5. Calculation of Cost Requirements

The calculation begins by determining the need for tools and materials to plan solar cell photovoltaics at the Al-Jihad Islamic Boarding School in Surabaya. After that, a price survey was carried out on the market. The unit price is based on the average price and the quality of the materials used in the planning. Good quality tools and materials can cause photovoltaic solar panels to experience damage or decrease efficiency more slowly than expected. The final calculation of the total planning budget is done by multiplying the requirements and the unit price of goods. After that, the total and the cost budget plan for the solar cell photovoltaic planning at the Al-Jihad Islamic Boarding School in Surabaya was obtained. The total budget calculation for solar cell photovoltaic planning is IDR. 209,850,400. Moreover, you can calculate the return on capital using the BEP (Break Even Point) formula.

BEP calculation is a position that explains expenditure and income at the same point. Therefore, the following calculations and equations are obtained:

Is known:

a) Total expenses = Rp. 209,850,400
b) PLN Watts = 12,000 Watts (12 KW)
c) Effective Hours = 4 hours/day (Joewono, et al. 2017)
d) Unit Price B-2/TR= Rp. 1,440.70/kWh (PLN, 2021)

Based on known data, the BEP can be calculated as follows:

BEP=(Rp.209,850,400)/(12 kW x 4 hours/day x Rp.1,440.70/kWh)

$$BEP = \frac{\text{Rp.209.850.400}}{12\text{ kW } x\ 4\ hour/\text{day } x\text{ Rp.1.440,70/kWh}}$$

$$BEP = \frac{\text{Rp.209.850.400}}{\text{Rp.69.153,6/day}}$$

$$BEP = 3.034{,}55\text{ days}$$

P-ISSN:2356-3109 E-ISSN: 2356-3117

If in 1 year there are 365 days, then it can be calculated as follows:

$BEP = 3.034{,}55 \text{ days } x \ \frac{1 \text{ year}}{365 \text{ days}}$

$BEP = 8{,}31 \text{ Years}$

The calculation results show that the BEP (Break Even Point) in planning solar cell photovoltaic at the Al-Jihad Islamic Boarding School in Surabaya is 8.31 years.

4. CONCLUSION

Based on the results and discussion, it can be concluded that the planning of photovoltaic solar cells at the Al-Jihad Islamic Boarding School in Surabaya was carried out based on electricity needs, and planning adjustments in selecting a 12 KVA inverter, 44 polycrystalline type photovoltaic solar cells, and other complementary materials such as cables, MCB and buffer. The amount of power produced from photovoltaic solar cells is 12 KWh with a construction cost of Rp. 209,850,400.,-. Meanwhile, efforts to mitigate climate change by reducing GHG emissions through this plan can reduce GHG from CO_2 1,200.5 kg, NH_4 3,013 kg, and N_2O 0.019 kg.

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